Verified Structured Problem-Solving Superiority and Obstruction-Certified Method-Language Expansion

ORION Paper X Ultimate-Successor Working Manuscript

August 2026

Abstract

Structured proof state, premise retrieval, proof repair, theorem proving, program synthesis, evolutionary code search, tactic learning, and tool-using language models can all improve difficult reasoning. ORION Paper X therefore targets a claim strictly above those donor mechanisms: across formal mathematics, algorithms/programs, scientific problem solving, and other verifier-backed domains, can a structured epistemic workspace improve verified solving under matched information and total compute, and, only when a prospectively frozen method basis is independently shown to be inadequate under the declared access/resource model, can ORION expand the method language with a minimal semantics-preserving edit that increases protected held-out reach without false escalation? Existing negative P9/P10 and ORION-Q results remain immutable, including regimes where exhaustive or donor-complete search already closes the registered space. P10-U gives first right of refusal to native proof-state systems, proof repair, failure-triggered learned interventions, tactic/macro/library mining, exact search, program synthesis, AlphaEvolve/CodeEvolve-style evolutionary method search, and P9 representation repairs. Method-language expansion is inaccessible until an Obstruction-Certified Method-Language Expansion (OCME) gate rules out representation, search, implementation, environment, library, and donor-search explanations under the same frozen model. A timeout is never an obstruction certificate, and a new tactic name or macro that expands to old closure is never method-space expansion. Primary outcomes include verified solve rate, verifier calls, branching/search depth, time-to-first-correct, invalid-action rate, repair success, obstruction-diagnosis accuracy, false expansion, outside-closure verification, held-out transfer, semantic preservation, and total resource cost. The intended terminal is replicated verified problem-solving superiority plus at least one independently verified outside-closure method-language expansion that transfers and remains unavailable to stronger search, synthesis, library learning, and evolutionary baselines at matched resources.

1 Largest scientific question

P10-U asks two nested questions:

1. Does a structured epistemic workspace improve *verified* problem solving under matched semantic information and total compute across heterogeneous domains?
2. When the frozen registered method language is genuinely inadequate, can ORION establish that inadequacy independently and then expand the language with a minimal semantics-preserving operator/grammar/interface edit that improves held-out verified reach?

The second question is not entered merely because the first system failed. Existing search, representation, proof-repair, synthesis, and evolutionary methods receive first right of refusal.

2 Nearest-work pressure moves the claim upward

Current systems already own much of the obvious method-search space. AlphaEvolve performs evaluator-guided evolutionary algorithm discovery over executable code [1]; CodeEvolve provides an open evolutionary coding framework for algorithm discovery [2]. In theorem proving, LAMP combines Lean 4, agentic coordination, MCP/tool access, and proof repair [3]. Learned interventions in Lean 4 `grind` activate learned search control only after stock automation fails, showing that failure-triggered learned repair itself is direct prior art [4].

P10-U therefore treats the following as donor-owned: proof-state conditioning, premise retrieval, proof repair, failure-triggered learned intervention, tactic/macro/library mining, exact/heuristic search, program synthesis, evolutionary program/method search, and P9 representation accessibility. ORION does not retreat to a tiny residual after absorbing these mechanisms. It moves upward to the relation

bounded obstruction $\rightarrow$ responsibility diagnosis $\rightarrow$ method-language opening $\rightarrow$ outside-closure edit $\rightarrow$ held-out v

The intended terminal remains **VERIFIED_PROBLEM_SOLVING_AND_METHOD_SPACE_EXPANSI**

3 Immutable negative and donor-closed history

Historical P10 structured-reasoning negatives, P9/P10 donor-subsumed results, and ORION-Q negatives are not relabeled. In particular:

1. if a finite registered method space is exhaustively searchable and exhaustive search matches ORION, no invention claim is permitted in that regime;
2. if a representation/interface change makes an existing method succeed, the result belongs to P9/accessibility rather than P10 method expansion;
3. if more search or a stronger registered prover closes the case, the result is search-limited rather than method-language-limited;
4. if a generated tactic merely abbreviates a composition already in the frozen closure, it is library/macro compression rather than expansion.

Issues #586 and #633 remain authority-bearing negative-result programmes for their covered P9/P10 and ORION-Q families. P10-U must beat those stronger explanations in a new regime rather than rewrite old outcomes.

4 Stage A: native-state verified problem solving

The first empirical gate is real verifier-backed solving. Compare under matched verifier-call, time, token, memory, and preprocessing budgets:

1. history/text only;
2. native proof state;
3. native state plus dependency/premise structure;
4. typed failure/repair state;

5. strong current prover/proof-repair baselines;
6. donor-complete state+search system;
7. ORION structured workspace.

Measure solve rate, verifier calls, branching factor, search depth, time-to-first-correct, invalid tactic rate, and repair success. Better tactic prediction without verified solve/search benefit is not sufficient.

The active #618 lane already implements a prospectively frozen native-Lean extraction and cross-revision programme, but its own authority contract states that no positive native-state result is claimed until its frozen analyzers execute and pass. P10-U preserves that boundary.

5 Stage B: action-abstraction phase diagram

Freeze action abstractions from atomic action through raw tactic, coarse family, effect-grounded action, macro/library action, and learned compositional action. The scientific question is when abstraction changes verified search efficiency or transfer. If a simpler old basis matches the richer system at matched resources, the richer mechanism is not credited.

6 Stage C: OCME obstruction certificate

Method-language expansion is inaccessible until all of the following are frozen and independently checked:

1. exact pre-freeze method basis and closure/composition operation;
2. exact computational, information, tool, and access model;
3. target task/family and verifier;
4. search/compute budget;
5. representation/interface repair routes from P9;
6. implementation/environment repair routes;
7. premise/library retrieval and macro-mining routes;
8. exact/heuristic theorem-proving baselines;
9. program-synthesis baselines;
10. evolutionary method/program-search baselines;
11. a bounded obstruction witness showing that the target remains outside registered reach under the declared model.

A timeout, low probability, failed proof attempt, or missing tactic prediction is not an obstruction certificate.

Valid obstruction evidence may be an exact finite non-reachability proof, a machine-checked lower bound, exhaustive bounded non-reachability, or an independently verified resource-bounded obstruction under a predeclared search class. If the registered access model is incomplete, the correct terminal is CannotCheck, not invention.

7 Stage D: method-language jump

Only an OCME-certified obstruction may open a method-language jump. Candidate edits may include:

1. a genuinely new semantic operator;
2. a new composition law unavailable in the old closure;
3. a new grammar/primitive that changes reachable verified solutions;
4. a representation-regime edit only when it changes method semantics/reach and carries an explicit correspondence;
5. a new verification/selection primitive that changes the reachable verified solution set.

A tactic alias or macro that expands into an existing old-language composition is classified as KnownComposition, not expansion.

8 Outside-closure and semantics verification

A claimed method-space expansion must satisfy all of:

1. an independent checker verifies that the candidate edit lies outside the frozen old closure;
2. the semantic target and verifier remain fixed, or an explicit valid correspondence is established;
3. no stronger oracle/tool/access channel is silently introduced;
4. the edit closes or materially moves the certified obstruction;
5. held-out tasks/families were frozen before edit generation and show positive transfer;
6. known-method controls show a low false-expansion rate;
7. strong search, synthesis, library-learning, and evolutionary baselines do not recover the same reach at matched cost;
8. an independent implementation/replay reproduces the reach expansion.

The generated method cannot certify its own novelty, correctness, or adoption authority.

9 Cross-domain programme

After formal non-vacuity, execute prospectively on:

1. algorithmic/program reasoning;
2. ORION-Q proof-carrying interface synthesis;
3. scientific problem reformulation/search/integration tasks from P1-U–P3-U;
4. historical reconstruction as mechanism-discovery evidence only;
5. at least one external verifier-backed domain with independent correctness authority.

A Lean-specific macro or local proof trick cannot support the broad terminal by itself.

10 Donor-complete baselines

The donor-complete comparator includes:

1. current theorem-proving/proof-state systems;
2. LAMP-like proof-repair/tool agents;
3. failure-triggered learned Lean interventions;
4. premise retrieval and dependency-aware proof search;
5. effect-grounded tactic models and tactic-library/macro mining;
6. generic LLM/tool agents;
7. exact and heuristic search;
8. program synthesis;
9. AlphaEvolve/CodeEvolve-style evolutionary method/program search;
10. P9 structured representation without P10 method edits;
11. search-more and compute-more controls;
12. representation-change-only controls;
13. oracle old-closure reach for exact synthetic worlds only.

All runnable comparisons receive the same evaluator, task information, access model, and total resource accounting.

11 Primary hypotheses

H1, verified problem-solving superiority. Across heterogeneous verifier-backed tasks, ORION achieves higher verified solve/success utility than the strongest runnable donor-complete comparator under matched information and total compute.

H2, search-efficiency superiority. ORION reduces verifier calls, search depth, branching, or time-to-first-correct at matched verified quality.

H3, obstruction-diagnosis validity. ORION distinguishes true method-language inadequacy from representation, search, implementation, environment, library, and verifier failures with lower false escalation than donor-complete diagnosis systems.

H4, outside-closure method-language expansion. At least one protected certified obstruction is closed by a minimal edit independently verified outside the old closure, with positive held-out transfer and no hidden access-model widening.

H5, low false expansion. On known-method controls, ORION rarely declares a language jump necessary when the registered basis suffices.

H6, cross-domain transfer. The problem-solving and OCME mechanism transfers prospectively to multiple domains without target-specific refitting.

12 Statistics and experimental units

The protected theorem/task is the default independent unit; repeated search seeds or model samples on one theorem are technical repeats unless a hierarchy is frozen. Method-expansion inference is reported by independently frozen obstruction family/task. Freeze family weighting, superiority/non-inferiority margins, confidence-interval method, multiplicity treatment, missing-run policy, and catastrophic semantic-failure rule before protected outcomes.

Report solve rate, verifier calls, search efficiency, false expansion, semantic failure, and held-out transfer separately. A mean solve-rate gain cannot compensate an invalid method-space claim.

13 Framework consistency

Current ORION already enforces the direction required by P10-U. The formal-engine readiness gate explicitly checks that ambiguous responsibility blocks revision, evidence/execution deficits are not reframing authority, and method change is protected rather than a local Reframe. It also distinguishes structural contract presence from independent formal closure. This supports P10's governance assumptions but does not itself provide a native Lean result or a method-basis obstruction certificate.

The following remain prospective rather than registered capabilities:

1. general method-basis obstruction certificates;
2. an outside-closure checker for arbitrary method languages;
3. autonomous OCME language jumps;
4. general method-space expansion superiority.

Repository-level integrity is not scientific authority. The hashes, frozen digests, deterministic replays and receipt chains this paper relies on establish that a stated computation was the computation performed and can be performed again. They establish nothing about whether its conclusion is correct, whether the benchmark measures what it claims, or whether an independent party has reviewed either. A perfectly replayed run of a flawed protocol replays the flaw exactly. Where this paper's claims require external scientific adjudication, that adjudication has not occurred and the integrity record does not substitute for it.

The synchronization verdict is **CONSISTENT_AS_PROSPECTIVE_EXTENSION**.

14 Adversarial controls

Mandatory attacks include:

1. search timeout mislabeled as method-language inadequacy;
2. generated edit merely renames/compresses known composition;
3. compiler/verifier feedback leaks the target action;

4. stronger oracle/tool/access model is silently introduced;
5. edit wins only on originating instances;
6. method invention fires when an existing method suffices;
7. native state improves prediction but not verified solve/search;
8. larger search budget closes the claimed ORION advantage;
9. representation change alone closes the obstruction;
10. proof-repair or learned-intervention baseline closes the obstruction;
11. donor-complete program synthesis or evolutionary search finds the same edit at matched cost.

15 Negative-result recursion as upward method-language search

Every negative is first localized to search complexity, representation accessibility, action abstraction, implementation/environment, verifier limitation, library/premise access, method-language inadequacy, donor domination, or non-identifiability. Only the method-language class opens OCME.

If a proposed jump is donor-owned, it is absorbed and the method language is generalized again. If the edit transfers poorly, the failure is retained and the causal responsibility is reopened. If materially distinct jump families saturate without a protected positive, P10-U seeks a lower bound or impossibility statement describing the capability required for a further jump rather than manufacturing novelty.

The detailed ChatGPT-led OCME programme is frozen under `research/claim_expansion/p10/gpt_r1/OCME_`

16 Claim ladder and current status

The intended progression is: native-state verified problem-solving gain; matched search-efficiency gain; valid obstruction diagnosis; one independently verified outside-closure method-language expansion; held-out transfer; cross-domain independent reproduction; **VERIFIED_PROBLEM_SOLVING_AND**

This TeX file is a prospective maximum-claim manuscript. It does not alter historical P10/P11 evidence, does not convert pending #618 executions into results, and does not assert method-space expansion before OCME's frozen gates are discharged.

References

[1] A. Novikov et al., "AlphaEvolve: A coding agent for scientific and algorithmic discovery." arXiv:2506.13131, 2025.

[2] H. Assumpção, D. Ferreira, L. Campos, and F. Murai, "CodeEvolve: An open source evolutionary coding agent for algorithm discovery and optimization." arXiv:2510.14150, 2025.

[3] S. Srinivasan R and M. Patawar, "LAMP: Lean-based Agentic framework with MCP and Proof Repair." arXiv:2606.28841, 2026.

[4] E. Wang, S. Chess, S. Szeto, and T. Meek, "Learned Interventions in Lean 4 grind." arXiv:2607.22972, 2026.